

CarMax Auto Owner Trust 2025-2

Observed 2026-08-31 | Valuation 2026-09-15 | Built 2026-10-02

Credit view

Central, Downside and Severe scenarios allocate \$0.00, \$0.00 and \$82,517,599.51 of terminal note principal impairment. Class protection and payment timing depend on the specified borrower and recovery assumptions. The opening collateral residual receives a separate adverse loss treatment; other source discrepancies remain unresolved. Market-price data is required before a relative-value investment recommendation.

Reverse stress for class D: first observed breach at default-hazard multiplier 2. The grid fixes adverse prepayment, recovery and collection assumptions. It does not establish a continuous minimum.

The analysis evaluates structural cash-flow resilience. An investment return conclusion requires an independent market price, executable spread and liquidity assessment. Discounted values in the workbook use editable yield and purchase-price assumptions.

Collateral and evidence

The case uses public SEC collateral disclosures and distribution reports. The latest observed pool is \$745.0m, versus an initial pool of \$1,410.6m. The latest active loan count is 47562. Source bytes, SHA-256 hashes and extraction provenance are preserved in the research pack.

The certificate exceeds mapped loan principal by \$512.01. Every scenario assigns this unreconciled amount immediate full loss with no recovery. This treatment does not bound other unresolved source cash differences.

Scenario comparison

Scenario	Pool net loss	Note loss	Unpaid interest
Central	\$19.4m	\$0.00	\$0.00
Downside	\$45.8m	\$0.00	\$0.00
Severe	\$124.4m	\$82.5m	\$17.6m

Class outcomes: Severe

Class	Opening	Protection	Loss	WAL (yrs)	Unpaid int.
A1	\$0.00	n.a.	\$0.00	n.a.	\$0.00
A2a	\$88.3m	83.1%	\$0.00	0.30	\$0.00
A2b	\$40.8m	83.1%	\$0.00	0.30	\$0.00
A3	\$452.1m	22.5%	\$0.00	1.75	\$0.00
A4	\$89.0m	10.5%	\$18.3m	n.a.	\$752,767.62
B	\$28.9m	6.6%	\$28.9m	n.a.	\$6.0m
C	\$21.2m	3.8%	\$21.2m	n.a.	\$5.9m
D	\$14.1m	1.9%	\$14.1m	n.a.	\$4.9m

What the model establishes

Historical reconciliation

Reported-data checks: 322 / 322 pass. Class interest matches: 128 / 128. Historical replay differences: 118. Replay uses observed cash and balances; it tests structural mechanics separately from borrower forecasting.

Calculated minus reported OC target: (\$76.08). Related payment differences remain visible. The strict tape/certificate gate has 287 unresolved comparisons; neither ingestion checks nor the small latest balance difference resolves them.

Borrower forecast evaluation

regularized multinomial logistic competing-event model with development-only calibration. Train: 2024-06-01 to 2025-06-30. Holdout: 2026-02-01 to 2026-08-31. Target: first-observed default disclosure, which can follow the effective default month. The same-originator unseen-deal test is separate from the later-month test.

Metric	Baseline	Fitted model	Unit
later months: observed-default brier	0.00317959	0.00252183	score
unseen deal: observed-default brier	0.00109394	0.000866513	score

Withheld servicing cash: unseen deal

Cash quantity / MAE	Baseline	Fitted model
gross defaults	\$607,571.74	\$463,350.53
principal collections	\$1.2m	\$4.5m
interest collections	\$153,077.97	\$121,736.91
ending pool	\$1.6m	\$8.2m

Across 7 withheld months, the fitted model improves default and interest errors but materially worsens principal collections and pool runoff. Scenarios are illustrative paths, not validated investment forecasts. The dashboard preserves both deals and monthly forecasts.

Assumptions and sensitivity

Severe: annual default rate: 12.3%; annual prepayment rate: 7.4%; recovery rate: 19.8%; recovery lag months: 7; interest collection factor: 0.85; floating note rate: 4.326%; discount rate: 8.0%. Credit scenarios require a Python rerun. Excel recalculates monthly discounted value, price per \$100, weighted-average life and balance roll-forward differences from exported cash flows.

Material limitations

- Conditional monthly hazards use observed beginning exposure for eligibility and dollar-error evaluation. They do not predict the undisclosed live fleet's eligibility.
- The later-month and unseen-deal tests are withheld from fitting and calibration; the two deals are from the same originator and do not establish other-originator generalization.

After-inspection study: transition default Brier beats fresh logistic in 4/4 folds.

Primary sources and reproducibility

Executed indenture: payment priorities (SEC)

Latest distribution certificate: observed balances and payments (SEC)